Term & Condition

1. Agreement and consent: By submitting this form, I acknowledge and agree to comply with the stated rules and regulations of the hackathon, ensuring the accuracy and truthfulness of the provided information. I affirm that the innovation being submitted for Tata Technologies InnoVent Hackathon is solely conceptualized by our team, and we hold the necessary rights and permissions to submit and present this idea. I understand and commit to adhere to the InnoVent Privacy Policy notice, Tata Technologies privacy policy and the terms and conditions outlined for participation. I also consent to the capture and storage of the provided contact details for communication and administrative purposes related to the hackathon.
   *

   InnoVent privacy notice: https://www.tatatechnologies.com/in/innovent-privacy-notice/Rules and regulations: https://www.tatatechnologies.com/in/tata-technologies-innovent-rules-and-regulations/Terms and conditions: https://www.tatatechnologies.com/in/tata-technologies-innovent-terms-and-conditions/Tata Technologies privacy policy: https://www.tatatechnologies.com/in/privacy-policy/

   ☑ I have read and agree to the InnoVent privacy notice, Rules and Regulations, Terms and Conditions and Tata Technologies Privacy Policy

Team Registration

2. Team Name *

   Add a creative and unique name that represents your team's identity and spirit.

   ForgeV

3. Team Leader Name *

   Enter the name of the designated team leader who will be the primary point of contact for the hackathon.

Suresh Kumar G

4. Team Leader Email Id *

This Email Id will be used for all further communication and updates related to the hackathon

sureshkumargane@gmail.com

5. Team Leader Gender *

Male

6. Team Leader Contact Number *

Provide the 10 digit contact number of the team leader for communication purposes during the hackathon.

9597727516

7. Select College Country (Country where your college is located)? *

India

8. College State (State where your college is located)? *

Tamil Nadu

9. Team Leader Institute Name *

Provide the name of the institute and the city

Madras Institute of Technology, Anna University, Chrompet, Chennai

10. College City (City where your college is located). *

Chrompet, Chennai

11. Team Leader Engineering Branch *

Select the engineering branch in which the team leader specializes from the dropdown list provided. If the branch is not listed, please choose "Other" and specify the branch in the provided text field.

Other

Production Engineering

12. Team Leader Graduating Year *

Select the anticipated graduating year of the team leader from the dropdown options

2028

13. Training & Placement Officer Name

Enter the name of the Training and Placement Officer

Enter your answer

14. Training & Placement Officer Email ID

Provide the email address for the Training and Placement Officer

Enter your answer

15. Faculty Advisor Name

Add the name of the faculty advisor who will be involved with your team

Dr. M. Manoj

16. Faculty Advisor Email ID

Provide the email address for the faculty advisor

manoj@mitindia.edu

17. Do you have team members? *

Yes

1st Team Member Registration

18. Team Member 1 Name *

Nitin Kalinga S V

19. Team Member 1 Email Id
*

nitinkalinga7@gmail.com

20. Team Member 1 Gender *

Male

21. Team Member 1 Contact number
*

7200177516

22. Team Member 1 Engineering Branch *

Other

Production Engineering

23. Team Member 1 Graduating Year *

2028

2nd Team Member Registration

24. Team Member 2 Name

Balaji Kumar M

25. Team Member 2 Email Id

balajikumar18.2006@gmail.com

26. Team Member 2 Gender

Male

27. Team Member 2 Contact Number

9445813903

28. Team Member 2 Engineering Branch

Other

Production Engineering

29. Team Member 2 Graduating Year

2028

3rd Team Member Registration

30. Team Member 3 Name

Jonathan Immanuel Kennedy

31. Team Member 3 Email Id

jonathanimmanuelkennedy@gmail.com

32. Team Member 3 Gender

Male

33. Team Member 3 Contact Number

7305299179

34. Team Member 3 Engineering Branch

Other

Production Engineering

35. Team Member 3 Graduating Year

2028

4th Team Member Registration

36. Team Member 4 Name

Akshan R

37. Team Member 4 Email Id

akshanathul@gmail.com

38. Team member 4 Gender

Male

39. Team Member 4 Contact Number

9840263566

40. Team Member 4 Engineering Branch

Other

Production Engineering

41. Team Member 4 Graduating Year

2028

Project details

42. Project Title *

Provide a concise title for your project that effectively captures its essence and purpose.

EdgeGuard AI - Vehicle Predictive Maintenance Simulation Platform

43. Please select project category under which you want to submit your project *

Select the project category to which your project belongs from the provided list of choices. You can refer to the rules and regulations for further details on each category.

- (x) AI at the Edge Solutions for Automotive
- ( ) AI at the Edge Solutions for Industrial Heavy Machinery
- ( ) AI at the Edge Solutions for Aerospace

44. Please select the project subcategory under the "AI at the Edge Solution for Automotive" category *

Edge AI for Vehicle Health and Predictive ...

45. Explain the specific real-world problem statement that your project is addressing *

In this field, please provide a clear and concise explanation of the specific real-world problem that your project aims to address. Describe the issue or challenge your project seeks to solve or improve upon, ensuring that it is well-defined and relevant to the chosen industry vertical and project category. Please share your response within 4000 characters.

The rapid adoption of electric vehicles (EVs) has increased the need for intelligent vehicle health monitoring and predictive maintenance systems. Most existing vehicles rely on periodic servicing or warning indicators that alert drivers only after a fault has already developed. This reactive maintenance approach can lead to unexpected vehicle breakdowns, higher repair costs, battery degradation, reduced driving range, safety risks, and increased vehicle downtime.

Additionally, current diagnostic systems primarily focus on reporting isolated faults and require cloud connectivity or specialized service tools for deeper analysis. Drivers often receive technical information that is difficult to understand, making it challenging for them to take preventive action. Important factors such as driving behavior, acceleration patterns, braking habits, traffic conditions, environmental temperature, charging practices, regenerative braking efficiency, and vehicle usage history are generally not analyzed together to provide personalized maintenance recommendations.

EdgeGuard AI addresses this challenge by introducing an intelligent Edge AI-based predictive maintenance platform that continuously monitors multiple vehicle subsystems locally on the vehicle. Instead of simply reporting failures, the system predicts future component health, identifies early signs of degradation, recommends preventive maintenance, and provides personalized suggestions based on the driver's real-world usage patterns. Since all AI inference runs entirely on the vehicle, the solution offers low latency, enhanced privacy, reduced cloud dependency, and uninterrupted operation even without internet connectivity.

This approach improves vehicle reliability, enhances battery lifespan, increases safety, minimizes maintenance costs, and provides drivers with an intuitive and user-friendly experience.

46. Explain how your project addresses the problem statement and the value it creates. *

Provide a detailed explanation of your proposed solution. Describe how it works, its key features, and the innovations involved, and explain how it addresses the problem statement. Highlight what makes your approach unique, practical, and effective. Please share your response within 4000 characters.

EdgeGuard AI is a comprehensive Edge AI-powered Vehicle Intelligence Platform that combines predictive maintenance, driver behavior analysis, and intelligent decision support into a single on-board system. The platform continuously analyzes vehicle parameters such as battery voltage, battery temperature, charging cycles, internal resistance, state of charge, ambient temperature, motor performance, braking efficiency, regenerative braking, and driving patterns. These inputs are processed locally using lightweight machine learning models without relying on cloud infrastructure. The system predicts the health of critical vehicle components including: Battery Motor Brakes Suspension Tires Cooling System Charging System Beyond component monitoring, EdgeGuard AI studies the driver's acceleration patterns, braking habits, average speed, daily travel routine, charging behavior, traffic conditions, weather conditions, and terrain characteristics to generate personalized recommendations that improve battery life and overall vehicle efficiency. The platform offers two dedicated user experiences: Driver View Simple and intuitive interface Easy-to-understand recommendations Battery health summary Remaining range prediction Charging suggestions AI maintenance alerts Insights View Advanced diagnostics Battery analytics Predictive health graphs Sensor visualization Feature importance analysis Technical maintenance insights The project also incorporates intelligent features such as AI-optimized charging recommendations, adaptive energy flow visualization, predictive service reminders, AI-assisted defect detection, and personalized eco-driving suggestions. By running all AI models directly on the vehicle, EdgeGuard AI delivers faster decision-making, improved data privacy, lower operational costs, and reliable performance without internet connectivity.

47. Explain the technologies/frameworks used in the above approach *

Elaborate on the technologies and frameworks employed in your project's approach. Describe the specific tools, software, programming languages, or frameworks utilized to develop and implement your solution. Highlight any innovative or notable technological components that contribute to the effectiveness and functionality of your project. Please share your response within 4000 characters.

The EdgeGuard AI prototype has been developed using modern software technologies that support Edge AI deployment, data visualization, and interactive user experience.

Programming Language
Python
AI & Machine Learning
Scikit-learn
XGBoost
NumPy
Pandas
Data Visualization
Plotly
Matplotlib
Frontend & Dashboard
Streamlit
HTML5

CSS3
JavaScript
UI Technologies
Custom Liquid Glass UI
Responsive layouts
SVG-based animated energy flow visualization
Dynamic vehicle dashboards
Interactive AI recommendation panels
Edge AI Characteristics
Local AI inference
No cloud dependency
Lightweight prediction models
Low-latency processing
Privacy-preserving architecture
Simulation Components
Driver behavior simulator
Battery health prediction engine
Charging optimization simulator
Vehicle health analytics engine
Predictive maintenance engine

The modular architecture allows additional AI models and vehicle subsystems to be integrated in future releases while maintaining efficient on-device execution.

48. Which AI category is your solution associated with? *

- [x] Edge AI
- [ ] Generative AI
- [ ] Edge AI Agentic AI
- [x] Predictive AI / Machine Learning
- [ ] Computer Vision / NLP / Speech AI
- [x] Hybrid AI / Combination of multiple AI approaches

49. Describe the key benefits and impact that your project aims to deliver. How will it improve or address the current situation? (In terms of numbers, if possible.) *

Highlight the key benefits and impact in quantifiable terms, which your project aims to bring to the current situation by addressing the identified problem. Please share your response within 4000 characters.

EdgeGuard AI aims to transform traditional reactive vehicle maintenance into proactive, AI-driven predictive maintenance.

Expected benefits include:

Up to 30% reduction in unexpected vehicle failures through early fault prediction.
15–20% improvement in battery lifespan by optimizing charging behavior and driving habits.
Up to 20% reduction in maintenance costs by identifying component degradation before major failures occur.
100% local AI processing, eliminating dependence on internet connectivity for core predictions.
Millisecond-level response time through Edge AI deployment.
Improved driver safety by providing early warnings for critical component issues.
Personalized driving recommendations that improve vehicle efficiency and energy utilization.
Better utilization of regenerative braking, contributing to increased driving range.
Simplified maintenance information for everyday users while providing detailed diagnostics for technicians.
Reduced service center visits by enabling preventive maintenance scheduling.

The platform also improves user confidence by presenting AI-generated recommendations in an intuitive and easy-to-understand interface. By combining predictive maintenance with personalized driver analytics, EdgeGuard AI creates a smarter, safer, and more efficient EV ownership experience.

50. What is the current state of your project? *

Select the current state of your project from the provided dropdown options. Ensure that you support your selection with relevant supporting artifacts and a project progress plan in your project presentation.

Proof of Concept Build in Progress

51. Does this project have Intellectual Property (IP) protection? *

No

52. If available, please mention the Patent Application Number, Patent Registration Number, Research Paper Title, DOI, Publication ID, or any other relevant reference

number.

Enter your answer

53. Please confirm if this innovation project is entirely conceptualized by your team from the beginning and is not a direct copy or replication of an existing innovation or technology? *

Please confirm the conceptualization of the project by the team and share the findings from a benchmarking study of recent advancements in the project domain by other OEMs or startups. Additionally, explain what differentiates your project from the existing available solutions. Kindly provide your response within 4000 characters.

Yes. EdgeGuard AI has been entirely conceptualized, designed, and developed by our team from the initial idea stage. The project is not a direct copy or replication of any existing commercial product.

As part of our research, we studied recent advancements in intelligent vehicle diagnostics and predictive maintenance adopted by leading automotive manufacturers and technology companies. While several manufacturers provide battery monitoring systems, cloud-based telematics, or isolated predictive maintenance features, most existing solutions are proprietary, hardware-specific, or focus on a limited set of vehicle parameters.

EdgeGuard AI differentiates itself through the integration of multiple intelligent capabilities within a unified Edge AI platform. Instead of monitoring only battery health, the system combines battery diagnostics, motor health, braking analysis, driver behavior modeling, charging optimization, environmental analysis, and predictive maintenance into a single on-device solution.

Another key differentiator is the dual-interface architecture. The Driver View presents simple, actionable recommendations suitable for everyday users, while the Insights View offers detailed technical analytics for engineers and service professionals. This dual-mode approach improves usability without sacrificing technical depth.

The solution also emphasizes privacy by processing all critical AI inference locally on the vehicle, reducing cloud dependency while enabling real-time decision-making. Its modular software architecture allows future integration of additional sensors, AI models, and vehicle subsystems, making it scalable and adaptable for next-generation connected electric vehicles.

54. Have you filed or submitted this project in any other contests, competitions, or events previously? If yes, please mention the name of events where you have demonstrated this project. *

Please mention if the project is presented in any other contest, competition, or events. Please share your response within 4000 characters.

No. This project has not been submitted or demonstrated in any previous contests, competitions, exhibitions, or innovation events. This submission represents the first public presentation of EdgeGuard AI.

Participant Confirmation

55. Please confirm whether you have entered all the necessary information in the form and are content with the provided responses *

Yes

56. How did you get to know about Tata Technologies InnoVent? *

Friends

57. Did you know about Tata Technologies as a company before participating in Tata Technologies InnoVent? *

- (●) Yes
- ( ) No

58. Add a Google Drive link to the folder with supporting documents: Detailed project presentation as per the template, supporting images and Videos of the POC and scanned copies of college ID card for all team members. *

Please provide a Google Drive link to the folder containing the supporting documents for your project with full-view access, detailed steps are shared in the Rules and Regulations (Point 4.4 and 4.5) for the same. The folder should include the detailed project presentation in the specified format as shared in the Section 4 of Rules and Regulations along with supporting images and videos showcasing the current state of POC as specified.
Refer Rules and regulations: https://www.tatatechnologies.com/tata-technologies-innovent-rules-and-regulations/

https://drive.google.com/drive/folders/1PSlFisGjKuZbJWC3gH1TMds4jWJUrc76?usp=drive_link